TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 8 OF 10

Balance and Swap



Big idea: Adjust one side, adjust the other. An equals sign means both sides are the same amount, like a balanced see-saw. If one number goes up, another goes down by the same amount to keep the balance.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Addition and Subtraction (The equals sign as 'is the same amount as' (signs + - =); Adding within 100; A missing number in a number sentence (blanks only, no letters)) Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'. Term 1 goes beyond the school Primary 1 syllabus on purpose: this lesson uses two-digit sums, tens and give-and-take, which the class explores with plates, tens bars and a worked partner step before anyone writes. The equals sign as balance is Primary 1. Compensation (giving and taking between two addends, as in 49 + 26 = 50 + 25) is beyond Primary 1 and sits at Think harder only. Letters for unknowns are Primary 6, so the lesson uses blanks only. The owner or a tutor should confirm that compensation placement.
Objectives	• Say that = means 'is the same amount as' and check whether two sums balance. • Find the missing number in a sentence such as 6 + 3 = 5 + ___ by thinking about how the numbers moved, not only by adding. • Know give and take: when one number goes up, the other goes down by the same amount and the total stays the same. • Use give and take to decide that two sums are equal without adding (Think harder), and to make an addition easier (49 + 26 = 50 + 25).
Joining cold	**Remember warm-up covers:** Bonds to 10 (how many more make 10); Bonds to 20 (how many more make 20, from 11 to 19); What = means: the same amount on both sides. Only counting to 20 and showing numbers with fingers. Nothing from the earlier Term 1 lessons is needed: the warm-up re-teaches bonds to 10 and 20 and the meaning of =, and the sums that follow use two plates and small things first. No carrying is ever needed. Not one of the lessons hardest to miss in Term 1 (Lessons 1 and 5 to 7 are), and Lessons 9 and 10 do not depend on it. A make-up is still recommended because Lesson 19 (Picture Equations) builds on balance. The safest lesson to skip in Term 1 is Lesson 4. A child who misses this one watches the recording and does workbook pages 1 and 2.
Child materials	• About 30 small things of one kind: buttons, dried pasta, paper clips or coins of one kind • Two plates, bowls or sheets of paper (one marked LEFT and one marked RIGHT) • The printed workbook (pages 1 to 5) and the kit page (tens bars and ones squares), already cut out by a grown-up • A pencil, a rubber and spare paper
Tutor materials	• A slide deck, and two plates with about 20 small things to show on camera if you wish • A timer slide or a visible clock No ruler is needed. Nothing needs to be bought: everything can be found at home. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears.
Tiers	Start here (7 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.

Contest-style move	Used at Think harder only: balance both sides of the equals sign; give and take (compensation) between two addends; say two sums are equal without adding. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, and try the two-plates routine with real things.
- Tell parents a week before: print workbook pages 1 to 6, cut out the kit page (page 6), find about 30 small things of one kind and two plates or bowls. Lesson 9 needs a centimetre ruler.
- Have the slides open and check the timer slide and the kit page are visible on screen share.
- Check the recording settings exclude children's video.

Safeguarding and recording (this lesson)

- Small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Sneak-peek and exit answers are shown on the child's own paper to the camera at the same moment, once. There is no private chat and no board held up in front of the group.
- Never read out a child's name with a right or wrong answer. Never show a child's page in the recording; record the tutor and the screen only, with children's video excluded.
- A child who ticks 'I am still stuck' gets a check-in in the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby). No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Balance and Swap	1:37
2–8	**Remember warm-up**	6	3: Warm-up: fingers to ten, Ten and more: how many to twenty?, The equals sign	4:37
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:33
12–28	**Explore**	16	3: Explore: two plates, Give and take, What did we notice?	12:16
28–38	**Mini-lesson**	10	7: Equals means balance, Give and take, Your turn: the missing number, Check: the missing number, Bigger numbers, same idea, Same total? Do not add yet, Quick-fire: is it balanced?	8:44
38–42	**Brain break**	4	1: See-saw stand-up	3:32
42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	25:06
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:05
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:49
82–85	**Look-back**	3	1: Look-back	2:28
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:38
	Total	**90**		**72:24**

The script needs 72:24 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 85:45. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: optional

items and extra quick questions first, then the second explore round, then any quick-fire in the mini-lesson, then the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:37 of 2:00

0:00	SLIDE	Show slide **Balance and Swap** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, two plates and your small things. Pause or rewind any time.*
0:36	SAY	Good morning, inquirers! Cameras on, pencil ready, two plates and your small things on the table. This lesson is recorded so that a friend who is away can catch up. We record me and the screen only. You may switch your camera off whenever you like.
0:59	SAY	Today is about balance, like a see-saw. A see-saw is level when both sides hold the same. Numbers can balance too. Let us find out how.
1:12	DO	Greet each child by first name and check their camera shows the page, the two plates and their hands. [0:25]
	NOTE	The routine and the four inquirer rules were taught in Lesson 1. Do not repeat them; a child who is new may hear them from you in one sentence: notice it, test it, explain why, mistakes help us.

2:00 – 8:00 Remember warm-up

script 4:37 of 6:00

2:00	SLIDE	Show slide **Warm-up: fingers to ten** (s-rem1)
2:25	REC	*Recording: join in! Show your fingers and say each answer out loud. Pause if you need longer.*
2:34	SAY	Fingers to ten. I say a number. Show me how many more fingers make ten.
2:41	ASK	Seven. How many more to make ten? [wait about 10 s for answers]
2:54	ASK	Four? [wait about 10 s for answers]
3:05	ASK	Nine? [wait about 10 s for answers]
3:16	ASK	Six? [wait about 10 s for answers]
3:26	ASK	Two? [wait about 10 s for answers]
3:36	SLIDE	Show slide **Ten and more: how many to twenty?** (s-rem2)
4:02	SAY	Now to twenty. Look at the full ten-frame and the extra counters. Thirteen is ten and three.
4:10	ASK	Thirteen. How many more make twenty? [wait about 12 s for answers]
4:25	ASK	Sixteen? [wait about 12 s for answers]
4:38	ASK	Eighteen? [wait about 12 s for answers]
4:50	ASK	Fifteen? [wait about 12 s for answers]
	IF STUCK	If a child hesitates on 16: point to the 6, ask how many more make ten (4), and say the other ten is already there.
5:02	SLIDE	Show slide **The equals sign** (s-rem3)
5:28	SAY	Now the equals sign. It does not mean the answer comes next. It means: is the same amount as. Three plus four is the same amount as five plus two.
5:42	ASK	Is three plus four the same amount as five plus two? Thumbs up or down. [wait about 12 s for answers]
6:02	ASK	Is three plus four the same amount as six plus two? [wait about 12 s for answers]
6:20	ASK	Is four plus four the same amount as five plus three? [wait about 12 s for answers]

	NOTE	Answers: 3, 6, 1, 4, 8 then 7, 4, 2, 5 then yes (7 and 7), no (7 and 8), yes (8 and 8). Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20.

8:00 – 12:00 Sneak-peek puzzles

script 3:33 of 4:00

8:00	SLIDE	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	REC	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	SAY	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	READ	Puzzle one. Ravi has 6 red beads and 4 blue beads. Siti has 5 red beads and 5 blue beads. Do they have the same number of beads? Draw dots to help, then circle Yes or No.
9:07	CHILDREN	Children draw dots or count and write yes or no. [0:50]
9:57	READ	Puzzle two. 8 plus 5 equals 9 plus a blank. What number goes in the blank? Write how you know.
10:08	CHILDREN	Children write or draw their answer. [0:55]
11:03	HOLD UP	Everyone holds their paper to the camera at the same moment, once, after puzzle two. Do not read answers aloud. No private chat.
11:28	SAY	Thank you. We will come back to puzzle two later.
	NOTE	Puzzle 1 answer: yes, 10 each. Puzzle 2 answer: 4. Look for good ideas to praise later (a child who drew the two sides, or who said 9 is one more so the blank is one less). Do not write scores or keep a list.

12:00 – 28:00 Explore

script 12:16 of 16:00

12:00	SLIDE	Show slide **Explore: two plates** (s-explore1)
12:25	REC	*Recording: pause now. Put your small things on the two plates and try each balance. Press play when you have tried.*
12:36	SAY	Today your hands are the teachers. Put your two plates side by side: left and right. On the left plate make a pile of six and a pile of four. Count them all.
12:52	SAY	On the right plate make a pile of five. Now make a second pile on the right plate so that both plates hold the same. How many are in your second pile?
13:08	CHILDREN	Round 1a: children build and count. Tutor scans the cameras and asks a few children to show their plates. [1:10]
14:18	ASK	How many in the second pile? How do you know the plates match? [wait about 30 s for answers]
14:54	SAY	Again. Left plate: seven and three. Right plate: eight and a second pile. Find the second pile.
15:03	CHILDREN	Round 1b: children build. Answer 2. [0:45]
15:48	SAY	One more. Left plate: nine and two. Right plate: eight and a second pile.
15:55	CHILDREN	Round 1c: children build. Answer 3. [0:45]
16:40	SLIDE	Show slide **Give and take** (s-explore2)
17:05	SAY	Round two: give and take. Left pile seven, right pile five. Count all the things on both piles. Now take one thing from the seven and give it to the five. Count all the things again.
17:23	CHILDREN	Children build 7 and 5, count 12, move one across (6 and 6), count again. [1:00]
18:23	ASK	Seven and five became six and six. Did the total change? [wait about 20 s for answers]
18:48	SAY	Do it again. Give one more across: five and seven. Again: four and eight. Count the total each time.
18:58	CHILDREN	Children give one across twice more and count the total each time. [0:50]
19:48	SAY	Now start again with seven and five. Give three across. What do you get? Count the total.
19:56	CHILDREN	Children try giving 3 across from 7 and 5 (4 and 8). The total is still 12. [0:40]

20:36	**SAY**	Round three. Left plate: eight and four. Right plate: nine and a second pile. The nine is one more than the eight. So what must the second pile be?
20:51	**CHILDREN**	Children build and test. Answer 3. [0:55]
21:46	**SAY**	One more. Left: six and five. Right: eight and a second pile. Eight is two more than six.
21:55	**CHILDREN**	Children build and test. Answer 3. [0:50]
22:45	**SLIDE**	Show slide **What did we notice?** (s-explore3)
23:10	**ASK**	What did you notice? When a number on one side got greater, what happened to the other number? [wait about 45 s for answers]
24:04	**SAY**	When one number goes up, the other goes down by the same amount. That keeps the balance. We call it give and take.
	IF STUCK	A child who changes both piles the same way (7 and 5 to 8 and 6): ask them to count all; is it still the same as before? Then take one back from the second pile.
	IF STUCK	A child with too few things: use the kit bars and squares, or fewer things: 6 and 4 against 5 and 5 is enough.
	NOTE	Main room only: no breakout rooms in this lesson, so you can see every camera. Building with the plates is for every child; the what-if card on the first explore slide is for any child who wants it, and nobody is named.

28:00 – 38:00 Mini-lesson

script 8:44 of 10:00

28:00	**SLIDE**	Show slide **Equals means balance** (s-m1)
28:25	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
28:32	**SAY**	Look. Five plus three on the left. Four plus four on the right. Both sides make eight. So the two sides balance, and we write an equals sign between them.
28:46	**ASK**	What does the equals sign mean? [wait about 15 s for answers]
29:04	**SLIDE**	Show slide **Give and take** (s-m2)
29:30	**SAY**	Here is give and take. Seven plus five is twelve. I take one from the seven, so it is six. I give that one to the five, so it is six. Six plus six. Still twelve.
29:48	**ASK**	If I take one from the seven, what must I do to the five? [wait about 18 s for answers]
30:12	**SLIDE**	Show slide **Your turn: the missing number** (s-m3)
30:38	**READ**	Your turn. Write the missing number on your paper: 8 plus 6 equals 7 plus a blank. 9 plus 4 equals 10 plus a blank. 6 plus 5 equals 4 plus a blank.
30:56	**CHILDREN**	Children write three numbers. [0:55]
31:50	**SLIDE**	Show slide **Check: the missing number** (s-m3a)
32:16	**READ**	Seven. Three. Seven.
32:17	**SLIDE**	Show slide **Bigger numbers, same idea** (s-m4)
32:42	**SAY**	Now bigger numbers. Twenty-five plus eighteen equals twenty-seven plus a blank. I do not need to add yet. The twenty-five became twenty-seven: two more. So the eighteen must go down by two.
32:58	**ASK**	What number is two less than eighteen? [wait about 15 s for answers]
33:17	**SAY**	Sixteen. Let us check. Twenty-five plus eighteen is forty-three. Twenty-seven plus sixteen is forty-three. It balances.
33:25	**SLIDE**	Show slide **Same total? Do not add yet** (s-m5)
33:50	**SAY**	Last idea. Are forty-seven plus eighteen and forty-five plus twenty the same? Do not add. Look at how the numbers moved.

34:00	**ASK**	Forty-seven went to forty-five. What happened to eighteen? Same total or different? [wait about 30 s for answers]
34:36	**SAY**	The 2 moved from one number to the other. So the totals are the same. Now look at the right side of the slide: twenty-nine plus fourteen and thirty plus fifteen. Both numbers went up, so the total went up.
34:56	**SLIDE**	Show slide **Quick-fire: is it balanced?** (s-m6)
35:21	**SAY**	Quick-fire. Is it balanced? Do not add. Look at how the numbers moved. Thumbs up for balanced, thumbs down for not.
35:32	**ASK**	Six plus five equals seven plus four? [wait about 15 s for answers]
35:50	**ASK**	Eight plus three equals nine plus three? [wait about 15 s for answers]
36:09	**ASK**	Twenty plus fourteen equals twenty-two plus twelve? [wait about 20 s for answers]
36:32	**SAY**	Well done for looking at how the numbers moved. When one goes up and the other goes down by the same, it balances.
	NOTE	Key words to use every time: balance, the same amount, give and take, blank. Do not teach 'compensation' or letters. Do not teach carrying. Totals are only used to check. Joining cold: every example can be shown with the two plates.
	IF STUCK	If a child writes the left total in the blank (8 plus 6 equals 7 plus 14): ask them to build both sides on the plates and say whether they match.

38:00 – 42:00 Brain break

script 3:32 of 4:00

38:00	**SLIDE**	Show slide **See-saw stand-up** (s-break)
38:25	**REC**	*Recording: stand up and join in! Arms out wide like a see-saw.*
38:31	**SAY**	Stand up and get a sip of water. Arms out wide like a see-saw. Left hand shows some fingers, right hand shows some fingers. I say two numbers that make ten, and you show them. Then I say give one across.
38:52	**CHILDREN**	Call 6 and 4, 5 and 5, 7 and 3. After each, say give one across and children move one finger to the other hand: still ten fingers. Then lean left, lean right, and sip water. [2:40]

42:00 – 70:00 Workbook

script 25:06 of 28:00

42:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
42:34	**READ**	Page 1, Start here. Question 1: fill in the blank so that both sides are the same. Question 2: Mei puts 6 buttons and 2 buttons on the left plate. Draw the right plate. Question 3: Siti writes 6 plus 3 equals 5 plus 9. Is it balanced? Then fill in the missing number.
43:03	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:00]
51:03	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
51:28	**READ**	Question 1: four, five, three, two. Question 2: one more button. Question 3: no, it is not balanced. The missing number is four.
51:41	**SAY**	Mark your own page in another colour. After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
51:54	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
52:18	**READ**	Page 2. Question 4: the total stays twelve. Question 5: look at how the first number moved. Question 6: Mei has 9 red pens and 6 blue pens. Ben has 10 red pens. Question 7: thirty-five plus twelve equals a blank plus two.
52:42	**CHILDREN**	Page 2, questions 4 to 7. [8:40]

61:22	SLIDE	Show slide **Check page 2** (s-ws2ans)
61:47	READ	Question 4: six, seven, eight. Question 5: ten, ten, four, nine. Question 6: Ben has five blue pens. Question 7: forty-five.
61:58	SLIDE	Show slide **Choose your next page** (s-choice)
62:23	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time.
62:36	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [4:30]
	IF STUCK	Q2 (drawing): draw each pile as dots and make the right plate match the left. Q3: build 6 and 3 on one plate, then 5 and some more on the other. Q5 and Q7: ask did the first number go up or down, and by how much? Q6: how many pens does Mei have all together?
	IF STUCK	A child who is still stuck ticks 'I am still stuck' on the look-back page. You check in during the open session in the next lesson (a quick visit to the shared room or a hand signal, parent nearby). Never a private message.
	NOTE	Think harder is for the idea taught today in a new setting: saying two sums are equal without adding. Go further is for mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree.

70:00 – 76:00 Talk time

script 5:05 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
70:32	READ	Ali has 6 plus 4, which is 10. He adds 1 to the 6 and adds 1 to the 4. He says 7 plus 5 is still 10. Do you agree? Use your plates to explain.
70:52	CHILDREN	Build 6 and 4, then 7 and 5 on the plates. Is the total still the same? [0:45]
71:37	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:55]
72:32	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 80 s for answers]
73:58	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
74:23	SAY	Remember puzzle two? Eight plus five equals nine plus a blank. Nine is one more than eight, so the blank is one less than five. The blank is four.
74:37	ASK	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: No. 6 + 4 is 10 but 7 + 5 is 12. Ali added 1 to both numbers, so the total went up by 2. To keep 10, add 1 to one number and take 1 from the other: 7 + 3. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:49 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	CHILDREN	Children answer E1 to E3 on their own. [2:50]
79:31	HOLD UP	Everyone holds their ticket to the camera at the same moment so only you see it. No private chat.
79:56	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:21	READ	Four. Yes. Two: the thirty went up by ten, so the twelve goes down by ten.
80:30	SAY	If you made a mistake, that is useful. Put a dot by it and tell your grown-up what you will try next time.

80:41	**SAY**	Check your own ticket in a different colour. Well done. I will stamp your inquiry journal.
	NOTE	Mark S, D or H in the tutor tracker (tutor only): S = secure on all three, D = developing (one wrong), H = needs help (two or three wrong). If D or H, plan two minutes of catch-up at the start of Lesson 9, in the open session, no names. Do not read out any child's result.

82:00 – 85:00 Look-back

script 2:28 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out.
82:48	**CHILDREN**	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 0:38 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Draw and Measure. You need a pencil, paper and a ruler marked in centimetres. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: what do you notice?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: twenty equals eleven plus a blank. And twenty equals twelve plus a blank. Keep going. What do you notice about the blanks?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 9, 8, 7, and so on; the first number goes up by one and the blank goes down by one. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Mei says: you can swap the two numbers in an addition and the total stays the same. Try 8 + 5 and 5 + 8, then 12 + 7 and 7 + 12, with your plates. Is it always true? What about taking away: 9 - 4 and 4 - 9?	*Recording: pause. Try 8 + 5 and 5 + 8 with your plates. Is swapping always true? Try 9 - 4 and 4 - 9.*	For adding, yes: the two piles just change places, so the total is the same. For taking away, no: 9 take away 4 is 5, but you cannot take 9 from 4.

Make a puzzle for your grown-up	4 (optional)	Pick a sum like 14 + 6. Change the first number and leave a blank for the second, like 16 + ___. Give it to your grown-up to fill in so that it balances.	*Recording: pause. Write a missing-number puzzle like 16 + ___ = 14 + 6 for a grown-up to solve.*	No single answer. Check by adding both sides: for example 14 + 6 = 20 and 16 + 4 = 20, so the blank is 4.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never seen a blank. Teach them how to find the blank in 25 + 18 = 27 + ___ without adding first.	*Recording: pause. Teach a grown-up how to find the blank in 25 + 18 = 27 + ___ without adding first.*	27 is 2 more than 25, so the blank is 2 less than 18. The blank is 16.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Ravi has 6 red beads and 4 blue beads. Siti has 5 red beads and 5 blue beads. Do they have the same number of beads? Draw dots to help, then circle Yes or No.

> **Answer: Yes, the same: 10 beads each.**

Puzzle 2 (think): 8 + 5 = 9 + ___. What number goes in the blank? Write how you know.

> **Answer: 4. 9 is 1 more than 8, so the blank is 1 less than 5. (Both sides make 13.)**

Unmarked. Papers are shown to the camera once, together. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Fill in the missing numbers.**

(a) 5 + 3 = 4 + ☐
(b) 6 + 4 = 5 + ☐
(c) 7 + 3 = ☐ + 7
(d) 2 + 8 = 8 + ☐

> **Answer: 4, 5, 3, 2**

Hint (say this first): Work out the left side first. Then find what the right side needs. You can use your two plates.
Worked solution: (a) 5 + 3 is 8 and 4 + 4 is 8. (b) 6 + 4 is 10 and 5 + 5 is 10. (c) 7 + 3 is 10 and 3 + 7 is 10. (d) 2 + 8 is 10 and 8 + 2 is 10.
Common mistakes: Writing the total of the left side in the blank (8 in part a).; Thinking the equals sign means the answer comes next.
Time for a middle child: about 3 min.

S2 **Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.**

She adds ☐ more buttons

> **Answer: 1 more button (6 + 2 = 8 and 7 + 1 = 8)**

Hint (say this first): Count the left plate: 6 and 2. Then draw 7 dots on the right and add dots until it matches.
Worked solution: The left plate has 6 + 2 = 8 buttons. The right plate has 7 buttons, so it needs 1 more to make 8.
Common mistakes: Drawing 8 more buttons on the right plate.; Forgetting the 7 buttons already there.
Time for a middle child: about 3 min.

S3 **Siti writes 6 + 3 = 5 + 9.**

(a) Is it balanced? Circle Yes or No.
(b) Fill in the missing number: 6 + 3 = 5 + ☐

> **Answer: No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next.**

Hint (say this first): Build 6 and 3 on your left plate. Then build 5 on the right plate. How many more make the plates match?
Worked solution: 6 + 3 is 9. The right side must also be 9. 5 and 4 more make 9, so the blank is 4. Siti wrote 9, the total of the left side, but the blank is only one of the two numbers on the right.
Common mistakes: Saying yes because 9 is the answer to 6 + 3.; Writing 9 in the box again (copying Siti).
Time for a middle child: about 2 min.

S4 **7 + 5 = 12. Fill in the missing numbers.**

(a) 6 + ☐ = 12
(b) 5 + ☐ = 12
(c) 4 + ☐ = 12

Answer: 6, 7, 8

Hint (say this first): The first number goes down by 1 each time, so the second number goes up by 1. You can use your plates.
Worked solution: (a) 6 + 6 = 12. (b) 5 + 7 = 12. (c) 4 + 8 = 12. The first number goes down by 1 and the second goes up by 1, so the total stays 12.
Common mistakes: Leaving the second number as 5.; Making the second number go down as well.
Time for a middle child: about 2.5 min.

S5 **Fill in the missing numbers.**

(a) 20 + 30 = 40 + ☐
(b) 50 + 20 = 60 + ☐
(c) 34 + 5 = 35 + ☐
(d) 61 + 8 = 60 + ☐

Answer: 10, 10, 4, 9

Hint (say this first): Did the first number go up or down? By how much? The other number does the opposite.
Worked solution: (a) 20 went up by 20 to 40, so 30 goes down by 20 to 10. (b) 50 went up by 10 to 60, so 20 goes down by 10 to 10. (c) 34 went up by 1 to 35, so 5 goes down to 4. (d) 61 went down by 1 to 60, so 8 goes up to 9.
Common mistakes: Writing the total of the left side (50, 70, 39 or 69).; Moving both numbers the same way (part c: 5 + 1 = 6).
Time for a middle child: about 3 min.

S6 **Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?**

Ben has ☐ blue pens.

Answer: 5 blue pens (9 + 6 = 10 + 5)

Hint (say this first): Count Mei's pens first. Then see how many more Ben needs after his 10 red pens.
Worked solution: Mei has 9 + 6 = 15 pens. Ben has 10 red pens, so he needs 5 more to make 15. Another way: his red pens are 1 more than Mei's, so his blue pens are 1 fewer: 5.
Common mistakes: Answering 6 (copying Mei's blue pens).; Answering 15 (the total).
Time for a middle child: about 3 min.

S7 **Fill in the missing number.**

35 + 12 = ☐ + 2

Answer: 45 (35 + 12 = 47 and 45 + 2 = 47)

Hint (say this first): Add 35 + 12 first. Then think: what number and 2 make that total? Or move a tens bar from the 12 to the 35.
Worked solution: The 12 went down by 10 to become 2, so the 35 goes up by 10 to become 45. Check: 35 + 12 = 47 and 45 + 2 = 47.
Common mistakes: Writing 47 (the total).; Writing 25 (taking a ten from the 35).
Time for a middle child: about 2.5 min.

Think harder (page 3): the taught idea in a new setting

T1 **Mei says 47 + 18 and 45 + 20 have the same total. Ali says 45 + 20 has the greater total. Do not add.**

(a) Who is right? Circle Mei or Ali.
(b) I know because ___.

Answer: Mei. 47 went down by 2 to 45 and 18 went up by 2 to 20. The 2 moved across, so both totals are 65.

Hint (say this first): Look at the first numbers: 47 and 45. Did it go up or down? By how much? What happened to 18 and 20?
Worked solution: 47 to 45 is 2 less. 18 to 20 is 2 more. The 2 moved from one number to the other, so the total stays the same. Both sums make 65.
Common mistakes: Saying 45 + 20 is bigger because 20 is a bigger number than 18.; Adding both sums and not saying how the numbers moved (correct, but check the reason).
Time for a middle child: about 5 min.

T2 **Do not add. Circle LEFT or RIGHT for the greater total, or circle SAME.**

(a) 29 + 14 and 30 + 13: LEFT / RIGHT / SAME
(b) 53 + 20 and 50 + 23: LEFT / RIGHT / SAME
(c) 26 + 15 and 28 + 16: LEFT / RIGHT / SAME

Answer: SAME, SAME, RIGHT

Hint (say this first): For each pair ask: did the first number go up or down, and by how much? Did the second number do the opposite, by the same amount?
Worked solution: (a) 29 up 1 to 30, 14 down 1 to 13: SAME (both 43). (b) 53 down 3 to 50, 20 up 3 to 23: SAME (both 73). (c) 26 up 2 to 28 and 15 up 1 to 16: both went up, so RIGHT is bigger (41 and 44).
Common mistakes: Saying SAME for part c because both numbers changed.; Looking only at the biggest numbers.
Time for a middle child: about 5 min.

T3 **Fill in the missing number. Then find the total.**

(a) 49 + 26 = 50 + ☐
(b) The total is ☐.

Answer: 25, then the total is 75

Hint (say this first): 49 is 1 less than 50. Give that 1 to the 26 or take it from the 26? Then add 50 and the new number.
Worked solution: 49 went up by 1 to 50, so 26 goes down by 1 to 25. 50 + 25 = 75, so 49 + 26 = 75.
Common mistakes: Writing 27 (making 26 go up as well).; Writing 75 in the blank.
Time for a middle child: about 4 min.

T4 **Ben writes 24 + 17 = 26 + 19.**

(a) Is it balanced? Circle Yes or No.
(b) Change the 19 to ☐.

Answer: No. 24 + 17 = 41 but 26 + 19 = 45. Both numbers went up by 2. Change 19 to 15.

Hint (say this first): Look at how 24 became 26, and how 17 became 19. Did the numbers move the same way, or opposite ways?
Worked solution: 24 went up by 2 to 26, so 17 should go down by 2 to 15. Ben made it go up by 2 to 19, so the right side is 4 too big. Check: 24 + 17 = 41 and 26 + 15 = 41.
Common mistakes: Saying it is balanced because both numbers changed by 2.; Changing the 26 and not saying which number you change.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Fill in the missing numbers so that all three sums have the same total.**

19 + 28 = 20 + ☐ = ☐ + 30

Answer: 27 and 17 (every sum is 47)

Hint (say this first): Find the first blank by giving and taking: 19 went up by 1. Then 20 + 27 = 47 is the total. What goes with 30?
Worked solution: 19 up 1 to 20, so 28 down 1 to 27. The total is 20 + 27 = 47 (check: 19 + 28 = 47). Then [] + 30 = 47, so the second blank is 47 - 30 = 17.
Common mistakes: Writing 27 in both blanks.; Writing 30 or 28 as the second blank.
Time for a middle child: about 5 min.

G2 **Fill in the blanks so both sides are the same. Write 3 different pairs. Many answers are right.**

(a) ☐ + 5 = 9 + ☐
(b) ☐ + 5 = 9 + ☐
(c) ☐ + 5 = 9 + ☐
(d) I notice ___.

> **Answer: Any pair where the first blank is 4 more than the second, for example 5 and 1, 6 and 2, 7 and 3 (also 4 and 0).**

Hint (say this first): Try a small number for the second blank, for example 1. What does the first blank need to be?
Worked solution: If the second blank is 1, the right side is 10, so the first blank is 5. If it is 2, the right side is 11, so the first blank is 6. The first blank is always 4 more than the second, because 9 is 4 more than 5.
Common mistakes: Writing the same number in both blanks.; Giving one pair only.
Time for a middle child: about 6 min.

G3 **Mei has 28 stickers on a page and 17 in her pocket. Ali, Ben and Kumar have the same total as Mei. Fill in the blanks.**

(a) Ali has 30 stickers on his page. His pocket has ☐ stickers.
(b) Ben has 25 stickers on his page. His pocket has ☐ stickers.
(c) Kumar has 12 stickers in his pocket. His page has ☐ stickers.

> **Answer: 15, 20 and 33 (Mei has 45 stickers; 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45)**

Hint (say this first): Ali's page has 2 more than Mei's page. So his pocket has 2 fewer. Ben's page has 3 fewer than Mei's page. For Kumar, his pocket has 5 fewer than Mei's, so his page has 5 more.
Worked solution: Mei has 28 + 17 = 45. Ali: 30 is 2 more than 28, so his pocket has 2 fewer than 17: 15. Ben: 25 is 3 fewer than 28, so his pocket has 3 more than 17: 20. Kumar: his pocket 12 is 5 fewer than 17, so his page has 5 more than 28: 33. Check: 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45.
Common mistakes: Writing 17 for Ali's pocket.; Making Ben's pocket 14 (moving the wrong way).; Writing 17 or 45 for Kumar's page.
Time for a middle child: about 7 min.

Investigation (page 5): Keep it the same

Easy first step: Add 6 + 4, then 7 + 3, then 7 + 5. Which total is not 10? Then look at 6 + 4 and 7 + 3: the 6 goes up 1. What does the 4 do?
Then: Pick two numbers. Add 3 to one and take 3 from the other. Then add 3 to both. Then take 2 from both. Which keep the total the same? By how much do the others change it?

> **Model answer: First step: 7 + 5 is not 10 (the totals are 10, 10 and 12). From 6 + 4 to 7 + 3 the 4 goes down 1. Next: add some to one number and take the same from the other and the total stays the same (give and take). Add 3 to both: the total goes up by 6. Take 2 from both: the total goes down by 4. When both numbers change the same way, the total changes by double.**

Look for: A child who tries more than one pair of numbers, says it stays the same when one goes up and the other goes down by the same amount, and notices that changing both numbers the same way changes the total by double. Any clear reason is fine.
Common mistakes: Saying adding the same to both numbers keeps the total the same (the total goes up by double).; Testing only one pair of numbers.

Talk and explain (minutes 70 to 76)

Ali has 6 + 4, which is 10. He adds 1 to the 6 and adds 1 to the 4. He says 7 + 5 is still 10. Do you agree? Use your plates to explain. Sentence starter: I think ... because ...

> **Model answer: No. 6 + 4 is 10 but 7 + 5 is 12. Ali made both numbers bigger, so the total went up by 2. To keep 10, add 1 to one number and take 1 from the other: 7 + 3.**

Look for: A reason that says one number must go up while the other goes down (not just 'because 12 is not 10'). Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Fill in the missing number.**
9 + 5 = 10 + ☐

Answer: 4

Worked solution: 9 went up by 1 to 10, so 5 goes down by 1 to 4. Both sides make 14.
Common mistakes: Writing 14 (the total).; Writing 6 (making the 5 go up).

E2 **Is 28 + 6 = 30 + 4 balanced?**
Circle Yes or No.

Answer: Yes

Worked solution: 28 went up by 2 to 30 and 6 went down by 2 to 4. Both sides make 34.
Common mistakes: Writing no because the numbers look different.

E3 **Fill in the missing number.**
30 + 12 = 40 + ☐

Answer: 2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2.

Worked solution: 30 went up by 10, so 12 goes down by 10 to 2. Both sides make 42.
Common mistakes: Writing 42 (the total).; Writing 22 (making the 12 go up).

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (knowing what = means; finding the blank; how the numbers moved; the big numbers (Think harder); something else); *What helped me* (two plates and small things; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my tutor will come and help)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Fill in the missing number: 6 + 3 = 5 + []	4
H2	Start here	Fill in the missing number: 10 + 4 = 9 + []	5
H3	Start here	Fill in the missing number: 20 + 30 = 40 + []	10
H4	Think harder	Ravi says 36 + 14 and 34 + 16 have the same total. Do not add. Is he right? Circle Yes or No. Then write: I know because ___.	Yes. 36 went down by 2 to 34 and 14 went up by 2 to 16. Both make 50.
H5	Go further	Put 12 small things in a pile of 7 and a pile of 5. Give 1 across and write the new sum, like 6 + 6 = 12. How many sums can you write?	Any list that shows the first number going down by 1 and the second going up by 1, for example 7 + 5, 6 + 6, 5 + 7, 4 + 8, 3 + 9, 2 + 10, 1 + 11, 0 + 12.

Online practice (`out/l08/practice`, key `t1l08`): 13 items; 7 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to see numbers as tens and ones and to say why. This week: the equals sign as balance, and finding a missing number by giving and taking.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Balance and Swap** s-title	TODAY WE WILL: / Balance two sides, and find missing numbers by giving and taking.	Lesson 8 of Term 1, 90 minutes. Objective: understand the equals sign as balance and find a missing number by giving and taking between two numbers. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Remind the group that the lesson is recorded (you and the screen only) and a child may keep the camera off.
2	2–8	**Warm-up: fingers to ten** s-rem1	[picture: ten_frame] / I show 7. How many more fingers make 10? / 7 4 9 6 2	Say each number; children show on their fingers how many more make ten. Answers: 7 needs 3, 4 needs 6, 9 needs 1, 6 needs 4, 2 needs 8. This is a spiral review of the bonds to 10 (holiday topic) and lets a new child join: they hear it before they say it. Hands up on camera is fine. No writing, no marks.
3	2–8	**Ten and more: how many to twenty?** s-rem2	[picture: ten_frame] / Start at 13. How many more make 20? / 13 16 18 15	Answers: 13 needs 7, 16 needs 4, 18 needs 2, 15 needs 5. Say: look at the ones part. 3 needs 7 to make ten, so 13 needs 7 to make twenty. Show two full ten-frames if a child is unsure. This is the bonds to 20 idea the lesson uses when one number moves to the next ten.
4	2–8	**The equals sign** s-rem3	LEFT SIDE: / 3 + 4 / [picture: ten_frame] / RIGHT SIDE: / 5 + 2 / [picture: ten_frame] / = means the same amount as / Is 3 + 4 the same amount as 6 + 2?	Answers: 3 + 4 is the same amount as 5 + 2 (both 7). 3 + 4 is not the same amount as 6 + 2 (7 and 8). 4 + 4 is the same amount as 5 + 3 (both 8). Say clearly that the equals sign does not mean the answer comes next; it means the two sides are the same amount. Children answer with a thumb up or down.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Ravi has 6 red beads and 4 blue beads. / Siti has 5 red beads and 5 blue beads. / Do they have the same number of beads? Draw dots to help, then circle Yes or No. / PUZZLE 2: think: / 8 + 5 = 9 + ___ / What number goes in the blank? Write how you know.	Puzzle 1 is an access question: any child can try it by drawing dots and counting. Puzzle 2 is a thinking question on the lesson's key idea. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: yes, 10 each; the blank is 4. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list of who found it easy.
6	12–28	**Explore: two plates** s-explore1	Put your two plates side by side: LEFT and RIGHT. / Left plate: a pile of 6 and a pile of 4. / Right plate: a pile of 5, and a second pile. / Make the plates match. How many in the second pile? / WHAT IF?: / What if the left plate has 7 and 3? What if it has 9 and 2?	Round 1, about 5 minutes. Objects first, no writing yet. Children build the left plate, count it, then build the right plate so that it matches. Ask a few children to show their plates to the camera. Answers: 5 (6 + 4 = 5 + 5), then 2 (7 + 3 = 8 + 2), then 3 (9 + 2 = 8 + 3). The what-if card is available from the start so nobody waits, and no child is named for choosing it. A child with too few things can use the kit squares.
7	12–28	**Give and take** s-explore2	Left pile 7, right pile 5. Count all. Move 1 across. Count all again. / 7 + 5 6 + 6 5 + 7 4 + 8 / Next, match 8 + 4 with 9 + a second pile.	Round 2 and round 3, about 8 minutes, main room. Children move one thing from one pile to the other and count the total again: it stays 12. Then give two or three across: 5 + 7 and 4 + 8 are still 12. Round 3 asks them to match 8 + 4 with 9 + a pile (3), and 6 + 5 with 8 + a pile (3), by thinking about how the first number moved. Walk the cameras and ask how do you know.

8	12–28	**What did we notice?** s-explore3	When one number went up, what happened to the other? / Did the total change when we gave one across? / How could you find the second pile without counting everything?	Take one or two answers from volunteers. Key idea: when one number goes up, the other goes down by the same amount, and the total stays the same. We call it give and take. Repeat a child's answer in your own words so the recording picks it up.
9	28–38	**Equals means balance** s-m1	LEFT SIDE: / 5 + 3 / [picture: ten_frame] / RIGHT SIDE: / 4 + 4 / [picture: ten_frame] / 5 + 3 = 4 + 4. Both sides make 8.	Show the two plates as a balanced see-saw. Both sides hold the same, so we write an equals sign between them. Ask: what does the equals sign mean? Answer: the two sides are the same amount. Say it is not a sign that the answer comes next.
10	28–38	**Give and take** s-m2	What we do | What we write / Start | 7 + 5 = 12 / Take 1 from the 7, give it to the 5 | 6 + 6 = 12 / Again | 5 + 7 = 12 / [picture: ten_frame] / [picture: ten_frame] / The total stays the same.	Think aloud: I take one from the seven, so seven becomes six. I give it to the five, so five becomes six. The total is still twelve. Ask: if I take one from the seven, what must I do to the five? Answer: give it one more. Link back to the Explore: this is what the plates showed.
11	28–38	**Your turn: the missing number** s-m3	Write on your paper. Timer: 1 minute / 8 + 6 = 7 + ___ 9 + 4 = 10 + ___ 6 + 5 = 4 + ___	Each child writes on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. Joining cold: they may use the plates.
12	28–38	**Check: the missing number** s-m3a	8 + 6 = 7 + 7 / 9 + 4 = 10 + 3 / 6 + 5 = 4 + 7	Answers: 7, 3, 7. Read each sentence aloud so a child watching the recording can mark their own work. The first number moved by 1 in the first two, and by 2 in the third: the blank moved the other way by the same amount.
13	28–38	**Bigger numbers, same idea** s-m4	25 + 18 = 27 + ___ / 25 went up to 27: 2 more. / So the blank is 2 less than 18. / The blank is 16. / Check: 25 + 18 = 43 and 27 + 16 = 43.	Worked example before the Start here page. Do not add first: look at how the number moved. 25 became 27, so 2 more. The other number must go down by 2: 18 becomes 16. Then check by adding both sides if the group is ready: both are 43. Ask: what number is two less than eighteen? Answer: sixteen.
14	28–38	**Same total? Do not add yet** s-m5	SAME TOTAL: / 47 + 18 and 45 + 20 / 47 goes down 2. 18 goes up 2. / NOT THE SAME: / 29 + 14 and 30 + 15 / 29 goes up 1. 14 goes up 1. / One number goes down and one goes up by the same amount. The total stays the same.	Worked example for Think harder. 47 to 45 is 2 less, and 18 to 20 is 2 more: the 2 moved across, so the totals are the same (65 each). 29 to 30 is 1 more and 14 to 15 is 1 more: both went up, so the total went up (43 and 45). Ask: same total or different, and how do you know? Keep to about two minutes; it is a worked example, not a skill to drill.
15	28–38	**Quick-fire: is it balanced?** s-m6	6 + 5 = 7 + 4 8 + 3 = 9 + 3 20 + 14 = 22 + 12 / Thumbs up: balanced. Thumbs down: not balanced.	Answers: yes (6 up 1 to 7 and 5 down 1 to 4), no (8 up 1 to 9 but 3 stays the same, so the right side is 1 too big), yes (20 up 2 to 22 and 14 down 2 to 12). Children answer with thumbs; nobody writes. Ask how do you know after each one, and repeat a volunteer's reason in your own words.
16	38–42	**See-saw stand-up** s-break	Arms out wide like a see-saw. Left hand and right hand show the fingers. / 6 and 4 5 and 5 7 and 3 / Give one across. Still ten fingers! Sip some water.	Four minutes. Stand up on camera. Call each pair that makes ten and have children show it on their two hands; then say give one across and they move one finger from one hand to the other. Count the fingers: still ten. Lean left, lean right, sip water. Keep it gentle for small spaces.
17	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Timer: 8 minutes / questions S1, S2, S3	Read questions 1 and 2 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder.
18	42–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group.

19	42–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 7. Timer: 9 minutes / questions S4, S5, S6, S7	Read the questions aloud. Question 6 is a story about pens: the children may draw the two sides. In question 7 the 12 became 2, which is one ten less: the other number goes up by one ten. Children may use the kit tens bars.
20	42–70	**Check page 2** s-ws2ans	answers S4, S5, S6, S7	Read each answer aloud so a child watching the recording can mark their own work. Question 6: Mei has 9 + 6 = 15 pens, and Ben has 10, so he has 5 blue pens. Question 7: 45, because 35 + 12 = 47 and 45 + 2 = 47. A child may find 47 first, or move a tens bar from the 12 to the 35. Never ask a child to show a wrong answer to the group.
21	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Same total without adding / Go further (page 4) | Two blanks and chains / Investigation (page 5) | Keep it the same / Check | Finish and check your own work / Everyone may choose. You can swap at any time.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 4 minutes.
22	70–76	**Talk and explain** s-talk	ALI SAYS: / 6 + 4 = 10. I add 1 to the 6 and 1 to the 4. / 7 + 5 is still 10. / Do you agree? Use your plates to explain. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. 6 + 4 is 10 but 7 + 5 is 12, because both numbers went up. To keep 10, add 1 to one number and take 1 from the other: 7 + 3. Repeat each child's answer in your own words for the recording.
23	70–76	**Back to the sneak-peek** s-sneakans	8 + 5 = 9 + ___ / The blank is 4. 9 is 1 more than 8, so the blank is 1 less than 5.	Reveal puzzle 2 now. Say: many of you wrote a good reason. Do not read out anyone's earlier answer. Both sides make 13.
24	76–82	**Exit ticket** s-exit	On your own. No looking back. Timer: 3 minutes / questions E1, E2, E3	Children answer on page 5. After the timer everyone holds the ticket to the camera at the same moment, so only the tutor sees it (no private chat). Stamp the inquiry journal.
25	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour.
26	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, check in during the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby); never a direct message. The Next time I will try row is optional; at the start of the next lesson you may say to the whole group: today you could try drawing it, using objects or checking your answer.
27	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Draw and Measure. Bring a pencil, paper and a cm ruler.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (stamps, the look-back). Do not start anything new. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
28	optional	**Go further: what do you notice?** s-opt	20 = 11 + ___ 20 = 12 + ___ 20 = 13 + ___ / Keep going. What do you notice about the blanks?	Optional, only if you are on time. Answers: 9, 8, 7, and so on. The first number goes up by 1 each time and the blank goes down by 1: give and take again. Take two answers and move on.

29	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Swap the two numbers in a sum. Does the total stay the same? / Make a puzzle | Write a missing-number puzzle for your grown-up / Teach it back | Show how to find the blank in 25 + 18 = 27 + ___ / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, two plates and your small things. Pause or rewind any time.*	Gather your workbook, pencil, paper, two plates and small things. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Show your fingers and say each answer out loud. Pause if you need longer.*	Play along. Show and say each answer before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Put your small things on the two plates and try each balance. Press play when you have tried.*	Pause. Build each balance on your two plates. Write what you find.
28:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Arms out wide like a see-saw.*	Stand up. Arms out like a see-saw and show the fingers.
42:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:37	1:52	ok
remember	6:00	4:37	5:28	ok
sneak	4:00	3:33	4:10	ok
explore	16:00	12:16	14:30	ok

mini	10:00	8:44	10:15	ok
break	4:00	3:32	4:11	ok
work	28:00	25:06	29:56	ok
talk	6:00	5:05	6:01	ok
exit	6:00	4:49	5:43	ok
look	3:00	2:28	2:55	ok
flex	5:00	0:38	0:45	ok
Total	90:00	72:24	85:45	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 7 questions; a middle child needs about 8.0 min for page 1 and 11.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.